

Car Price Prediction Project

A project report submitted in training program on

Debasish Sahoo

Artificial Intelligence – Machine Learning

Radhika Maggo

23071003513

(2023-2026)

Dr. Virendra Swarup Institute Of Computer Studies

CANDIDATE’S DECLARATION

Radhika Maggo, hereby certify that the work presented in this report, titled __***Car Price Prediction***__, submitted in partial flment of the requirements for the training programme on Artificial Intelligence – Machine Learning, is an authentic record of my own work. I have also duly cited all references for any data, text(s), figure(s), table(s), or equation(s) that have been taken from other sources.

Date:

1/07/2025

Signature of the Candidate

Radhika Maggo

ABSTRACT

The **Car Price Prediction Project** aims to develop a machine learning model that accurately predicts the selling price of a car based on various attributes such as year, present price, kilometers driven, fuel type, seller type, transmission, and ownership. The project involves data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and performance evaluation.

Key steps include cleaning and transforming the data, encoding categorical variables, and using correlation analysis to select important features.

Multiple regression algorithms such as Linear Regression, Lasso Regression, and Ridge Regression are implemented and compared to identify the best-performing model. The final model demonstrates strong predictive accuracy and can be used to support car valuation processes in the automotive and resale sectors.

Table of Contents

1. **Introduction**
 - Overview of the project and objectives
2. **Importing Libraries**
3. Loading necessary Python libraries (e.g., pandas, matplotlib, seaborn, sklearn)
4. **Loading the Dataset**
 - Reading the car data from CSV file
5. **Initial Data Exploration**
 - Displaying the first few rows
 - Data types and basic statistics
 -
6. **Data Cleaning and Preprocessing**
 - Checking for null or missing values
 - Handling categorical variables (e.g., Fuel_Type, Seller_Type, Transmission)
 - Dropping unneeded columns (like 'Car_Name')
 -
7. **Feature Engineering**
 - Creating a new feature for car age
 - Dropping the 'Year' column

8. Exploratory Data Analysis (EDA)

- Pairplots and heatmaps
- Correlation analysis to identify important features

9. Feature Selection

- Defining dependent and independent variables
- Using SelectKBest or correlation-based methods (if applicable)

10. Train-Test Split

- Splitting the data into training and test sets

11. Model Building and Training

- Linear Regression
- Ridge Regression
- Lasso Regression

12. Model Evaluation

- R^2 Score
- Mean Absolute Error (MAE)
- Visualizing predicted vs actual prices

13. Model Comparison

- Comparing performance of different regression models

14. Model Deployment (Optional/Placeholder)

- Notes on how the model can be deployed using tools like Flask or Streamlit (if included)

15. Conclusion

- Summary of findings
- Best performing model and its accuracy

16. References & Acknowledgements

- (If applicable)

Introduction

In today's data-driven automotive market, accurately predicting the resale price of a used car is both a complex and valuable task. The car price is influenced by various factors such as the vehicle's age, brand, fuel type, kilometers driven, ownership history, and more. The objective of this project is to develop a predictive machine learning model that estimates the selling price of a used car based on its key features.

This project follows a structured machine learning pipeline including data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model selection, training, and evaluation. We apply multiple regression algorithms—such as Linear Regression, Ridge Regression, and Lasso Regression—to find the best-performing model in terms of accuracy and robustness.

By the end of this project, we aim to:

- Understand the key factors affecting car prices
- Build an accurate regression model to predict prices
- Provide a foundation for real-world deployment (e.g., car resale platforms)

This model can be a valuable tool for car dealerships, online marketplaces, and individual sellers or buyers looking to make informed pricing decisions.

References

1. Kaggle Dataset – Used Car Dataset

- Original dataset or similar datasets often used in car price prediction projects are sourced from Kaggle.
- <https://www.kaggle.com/datasets>

2. Scikit-learn Documentation

- Official documentation for machine learning models, preprocessing tools, and metrics used in this project.
- <https://scikit-learn.org/stable/documentation.html>

3. Pandas Documentation

- For data manipulation and analysis.
- <https://pandas.pydata.org/docs/>

4. Matplotlib & Seaborn Documentation

- Used for data visualization in EDA.
- Matplotlib: <https://matplotlib.org/stable/contents.html>
- Seaborn: <https://seaborn.pydata.org/>

5. Python Official Documentation

- Reference for base Python functions and data structures.
- <https://docs.python.org/3/>

6. Machine Learning Books & Courses

- Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
- Andrew Ng's Machine Learning Course on Coursera: <https://www.coursera.org/learn/machine-learning>

Quick Reference

Step	Description
Data Import	<code>pd.read_csv()</code> to load the dataset
Data Cleaning	Remove unnecessary columns, handle missing values, encode categorical data
Feature Engineering	Create new features like <code>Car_Age</code> , drop redundant ones like <code>Year</code> , <code>Car_Name</code>
EDA	Use <code>seaborn</code> , <code>matplotlib</code> for correlation heatmaps, pairplots, and insights
Feature Selection	Analyze correlation or use model-based feature importance
Train-Test Split	<code>train_test_split(X, y, test_size=0.2)</code>
Model Building	Apply Linear, Ridge, and Lasso Regression models using <code>sklearn.linear_model</code>
Evaluation Metrics	<code>r2_score</code> , <code>mean_absolute_error</code> , <code>plt.scatter(y_test, predictions)</code>
Best Model	Select model with highest R^2 and lowest MAE on test data
Deployment (Optional)	Can be done using <code>Flask</code> , <code>Streamlit</code> , or <code>Pickle</code> for saving the model

Car Price Prediction Analysis Overview

Radhika Maggo
Artificial Intelligence\$ machine Learning



Car Price Prediction Introduction



01

Market Analysis

Helps investors understand the market trends and fluctuations in car prices for informed decisions.

02

Insurance Evaluation

Assists insurance companies in determining accurate premiums based on predicted vehicle values.

03

Loan Assessment

Facilitates banks and lenders in evaluating the amount of credit to provide based on predicted car prices.

04

Consumer Insights

Empowers consumers with knowledge on fair pricing, helping them negotiate better deals while purchasing.

Importance of Car Price Prediction

Consumer Insights

Provides valuable data on buyer preferences and spending patterns.

Competitive Edge

Enables businesses to remain competitive in pricing strategies.

Investment Decisions

Guides investors in making informed purchasing or selling strategies.

Risk Management

Assists in identifying potential risks and preventing losses.

Market Trends

Understanding market shifts helps strategize car pricing effectively.

Sales Forecasting

Aids in predicting future sales based on price dynamics.



Data Sources for Car Pricing

	Average Price	Max Price	Min Price	Transaction Count
Year	2021	\$30,000	\$40,000	\$20,000
Model	2022	\$28,000	\$38,000	\$18,000
Condition	2023	\$32,000	\$42,000	\$22,000
Mileage	2024	\$35,000	\$45,000	\$25,000

Insights

01

Pricing Trend

Prices increased 20% over last year.

02

High Mileage Impact

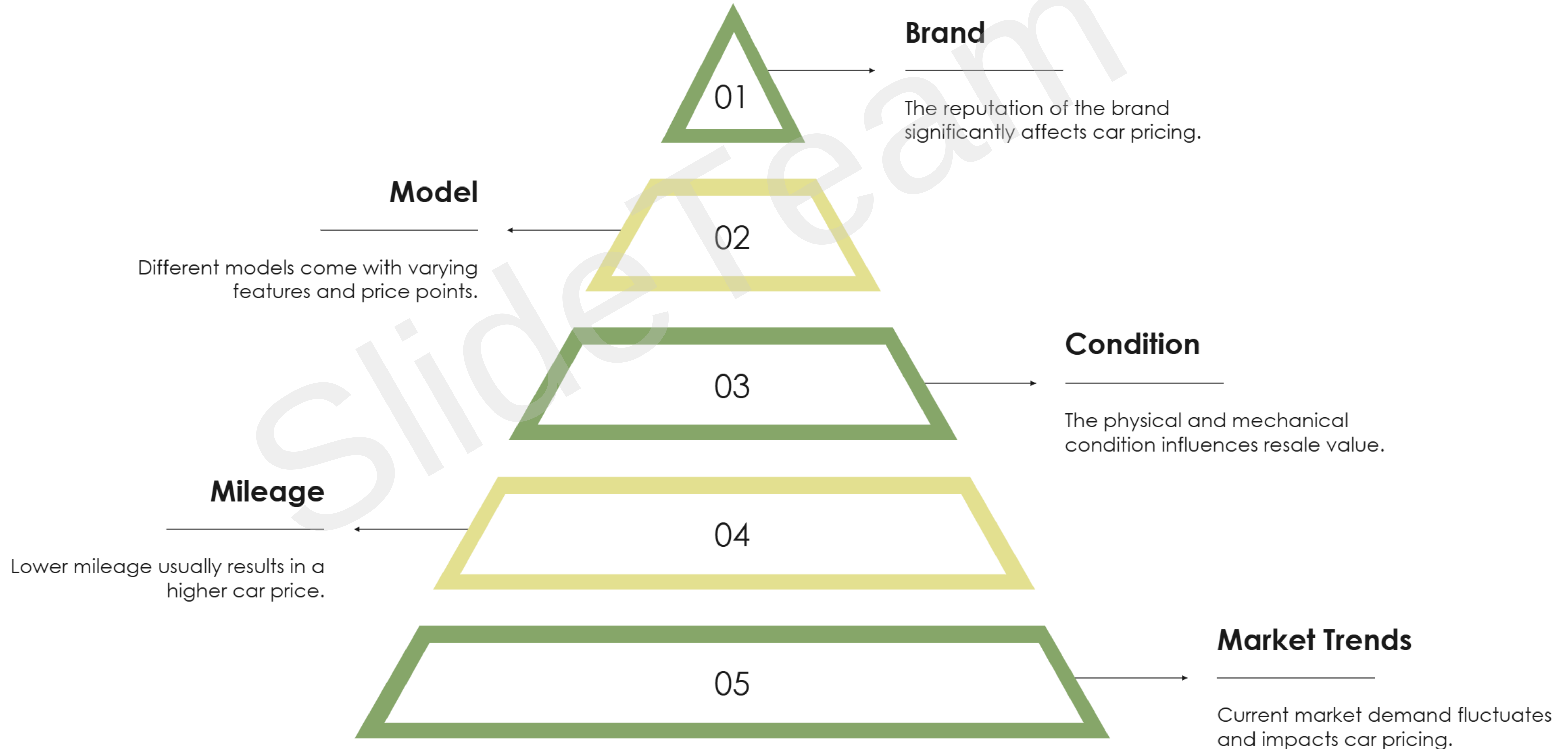
High mileage decreases value by 30%.

03

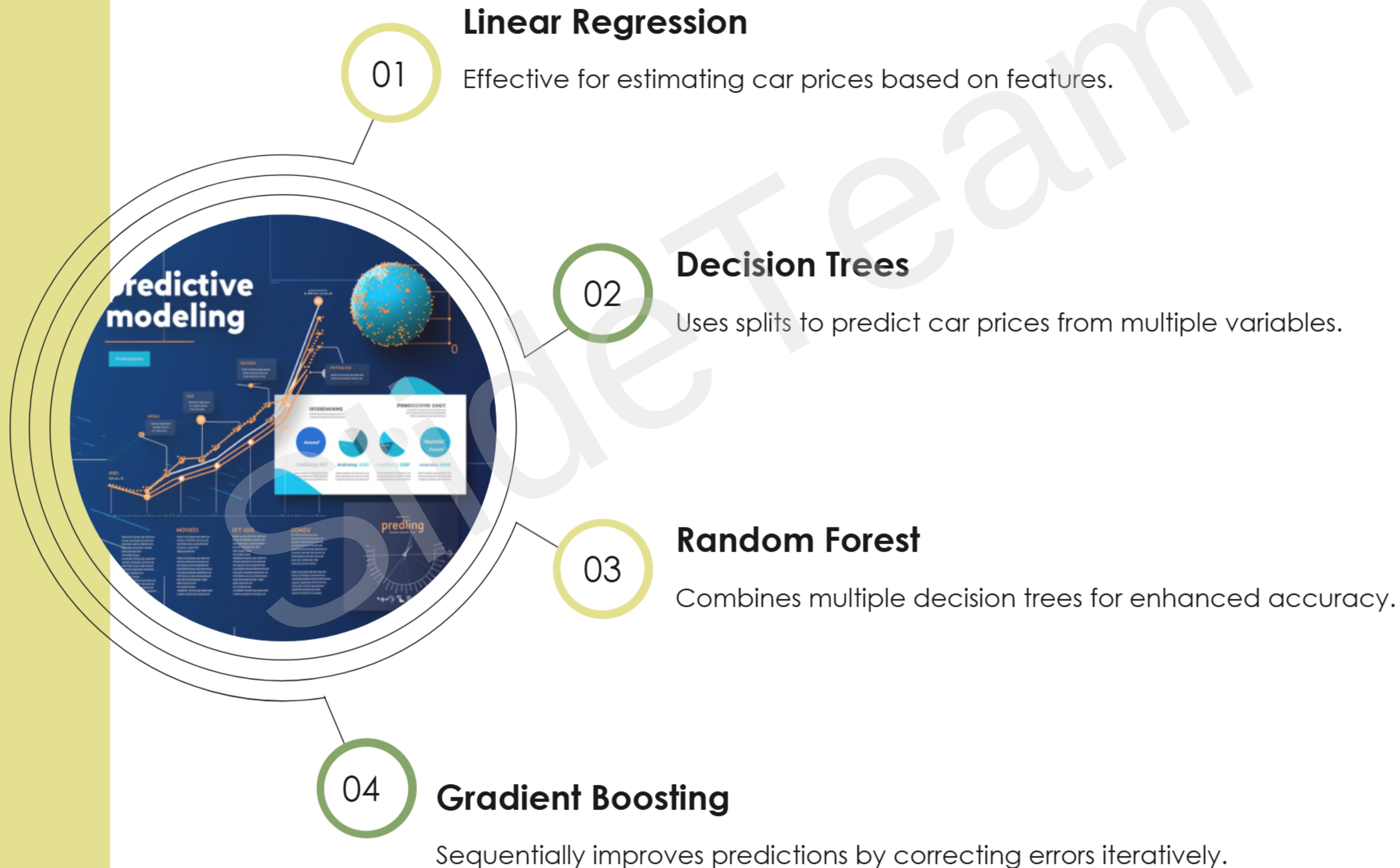
Luxury Model Demand

Luxury models are 15% more sought after.

Factors Influencing Car Prices



Machine Learning Models Overview



Evaluation Metrics for Predictions

MAE

180



MSE

90000



R^2

0.85

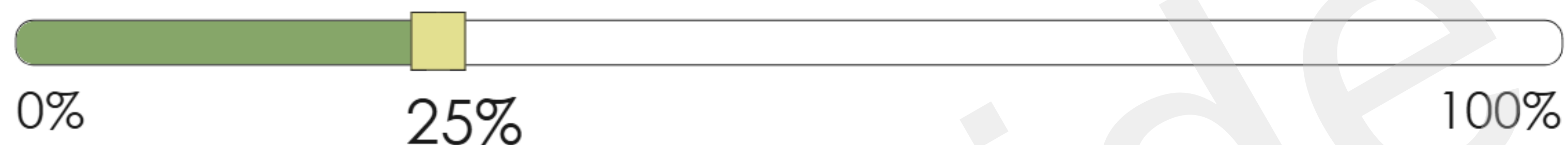


MAPE

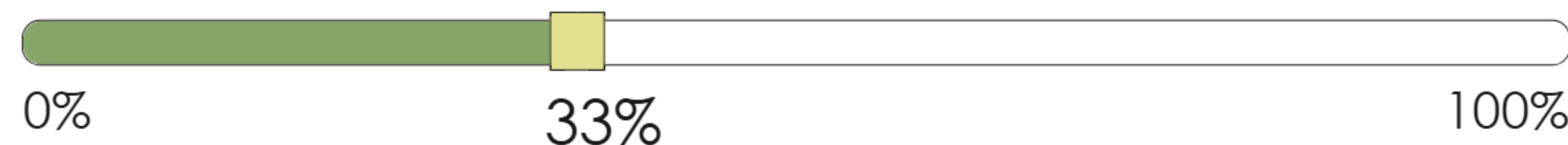
5%



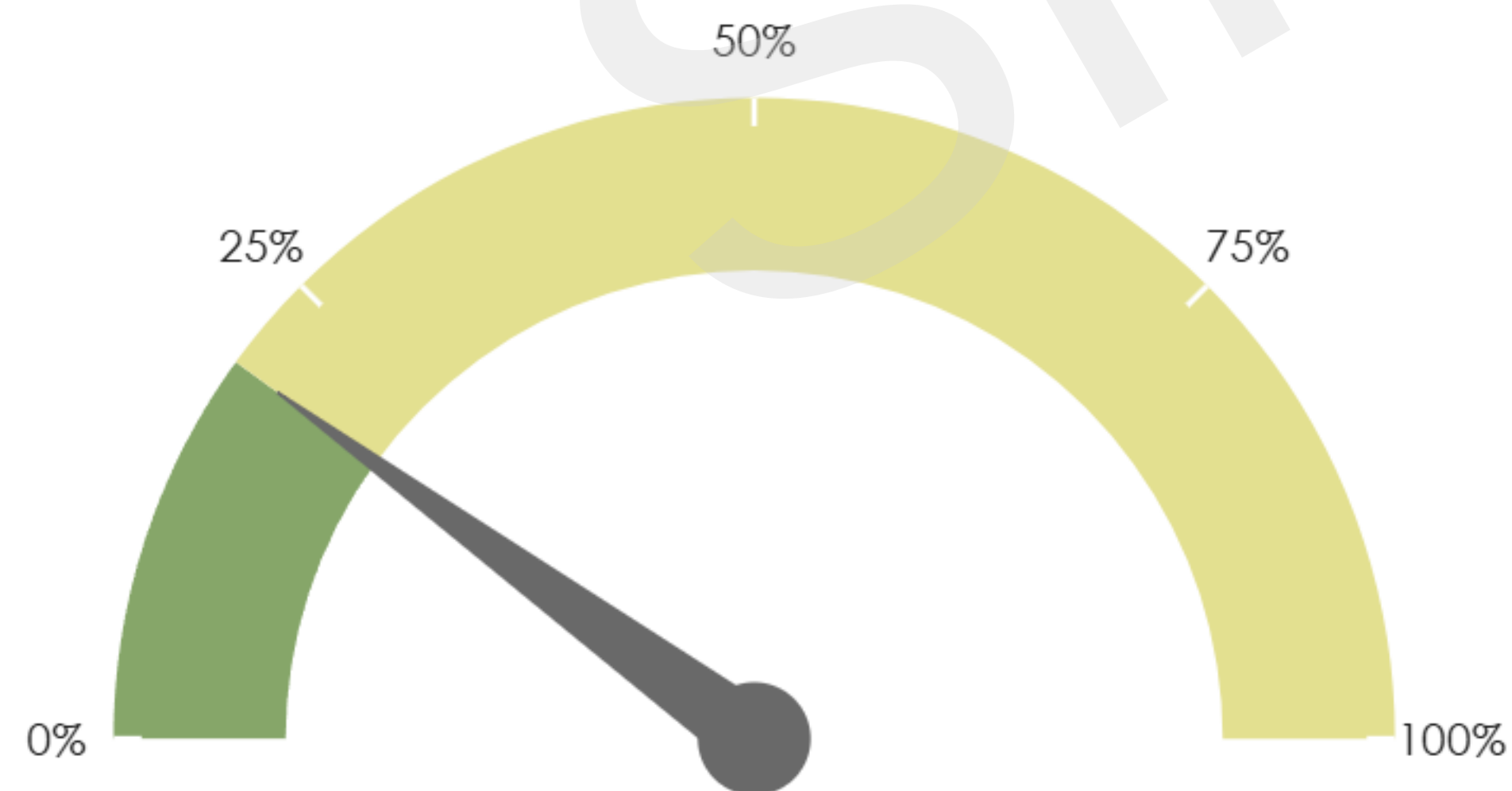
Model Accuracy



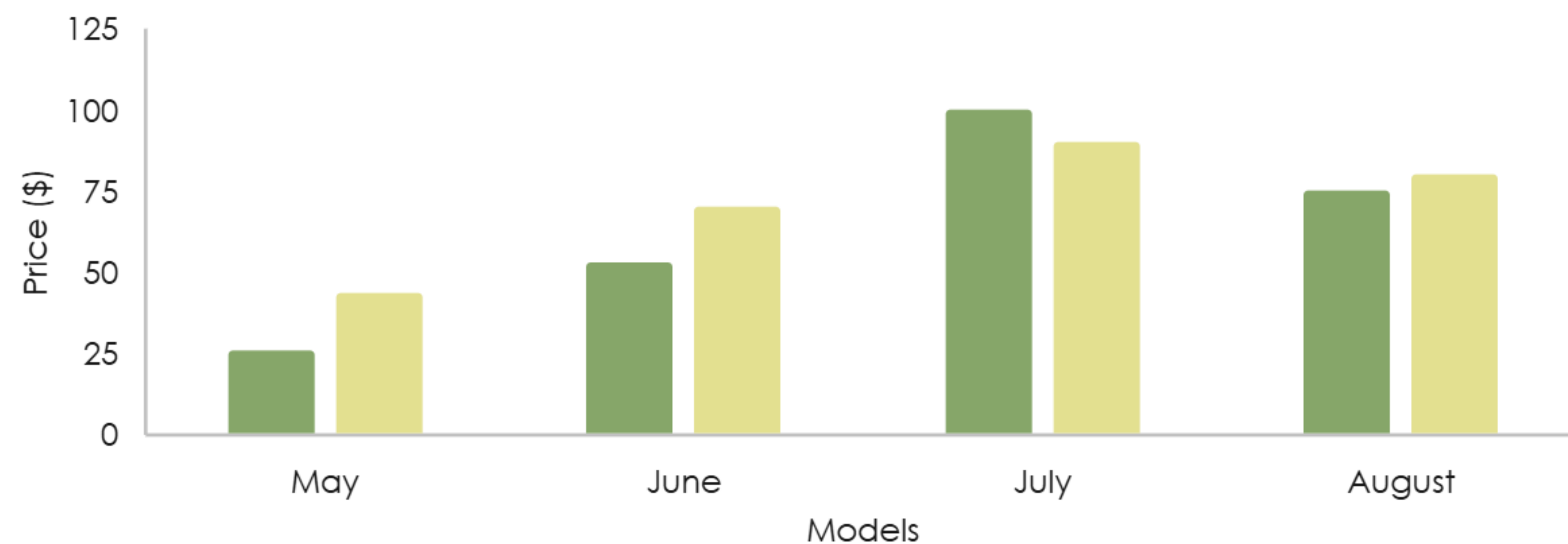
Prediction Error Margin



Model Performance Speedometer



Comparison of Predicted Prices



This is a sample dashboard and is 100% editable. Please edit the metrics according to your message.

Real-World Applications of Predictions



Problem Faced

Accurate pricing can be challenging for buyers.



Solution Offered

Utilize machine learning for price prediction.



Benefits

Enhances decision-making for both buyers and sellers.

Approach

01

Data

Collect historical price data and features.

02

Model

Select suitable predictive modeling techniques.

03

Training

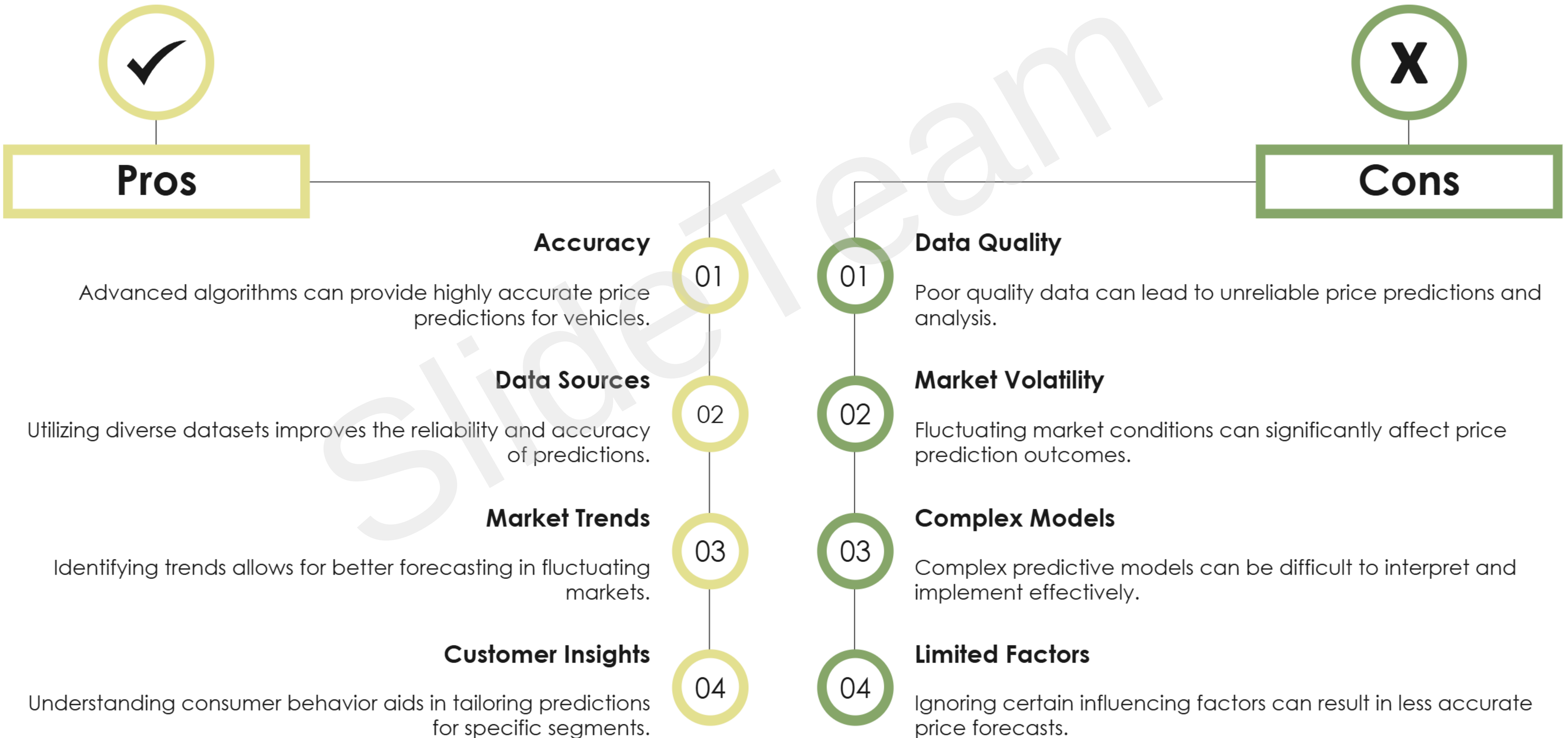
Train the model using the collected data.

04

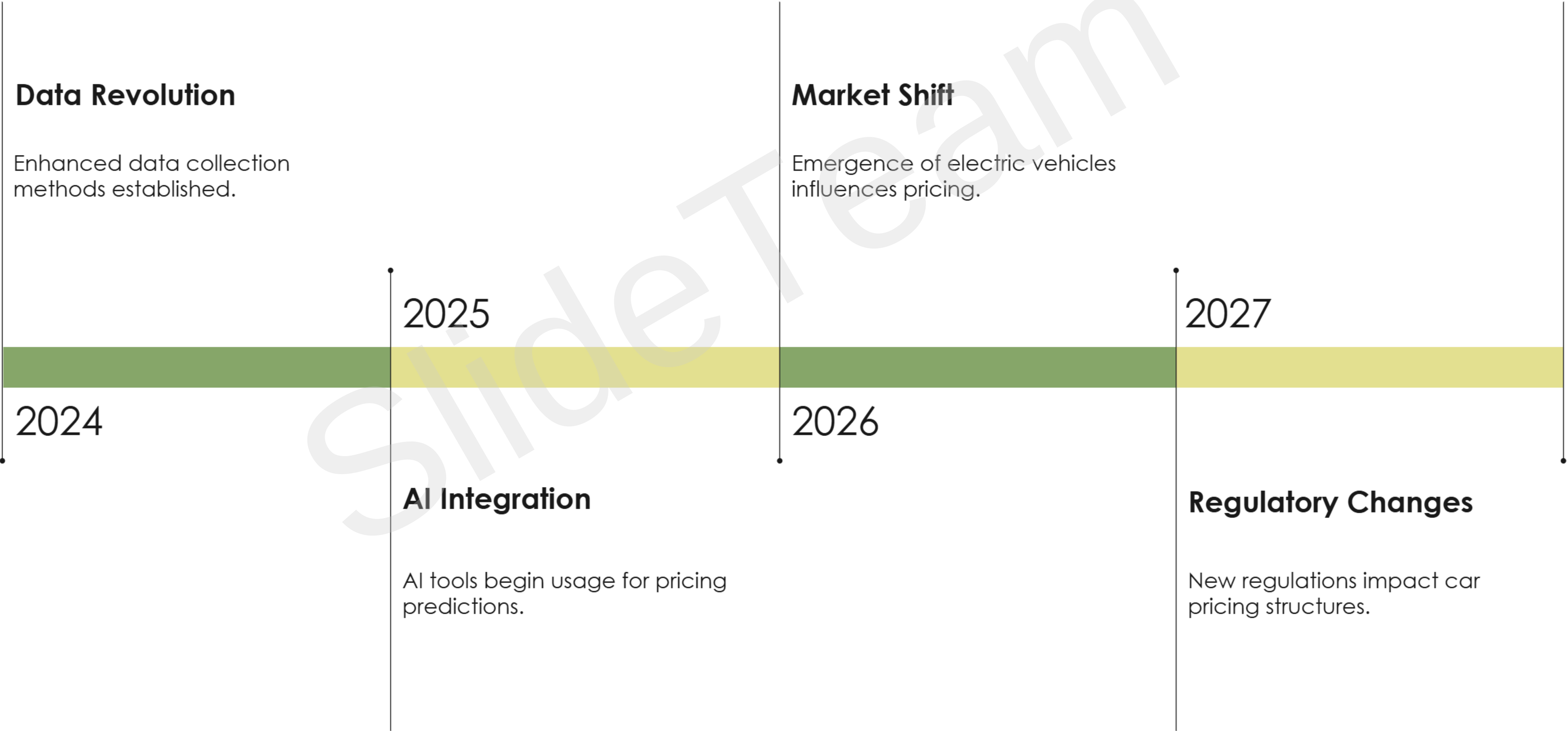
Evaluation

Assess model performance with test data.

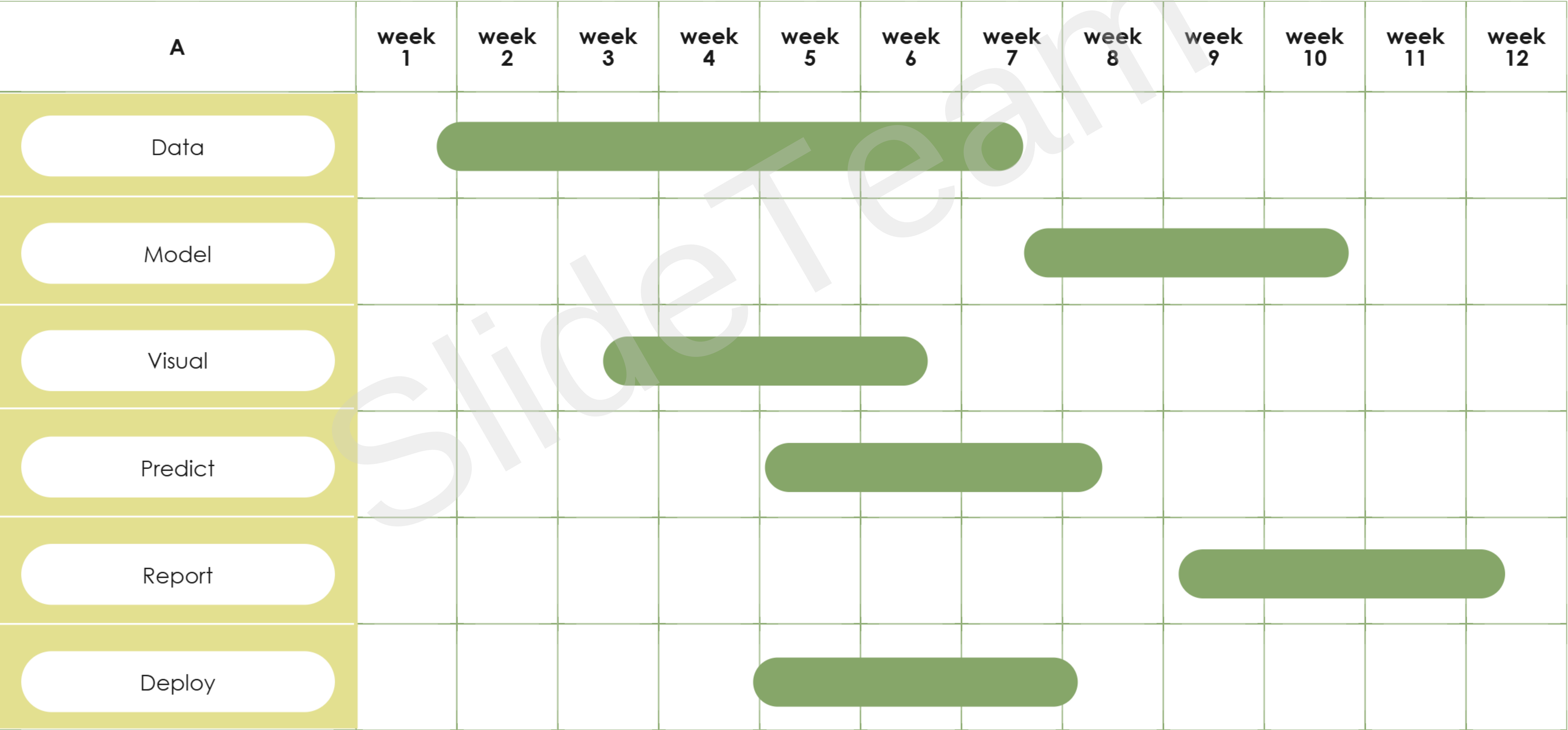
Challenges in Car Price Prediction



Future Trends in Car Pricing Models



Tools for Data Analysis and Visualization



This is a sample Gantt Chart. Please edit the timeframe above according to your schedule.

Case Studies of Successful Predictions

Problem Faced

Inaccurate price estimation for used cars.



Solution Offered

Implemented machine learning algorithms for predictions.



Benefits

Enhanced pricing accuracy leading to better sales.



User Feedback and Testimonials



John



“

This car price prediction tool saved me a lot of money.

”



Sara



“

I was impressed by the accuracy of the predictions provided.

”



Mike

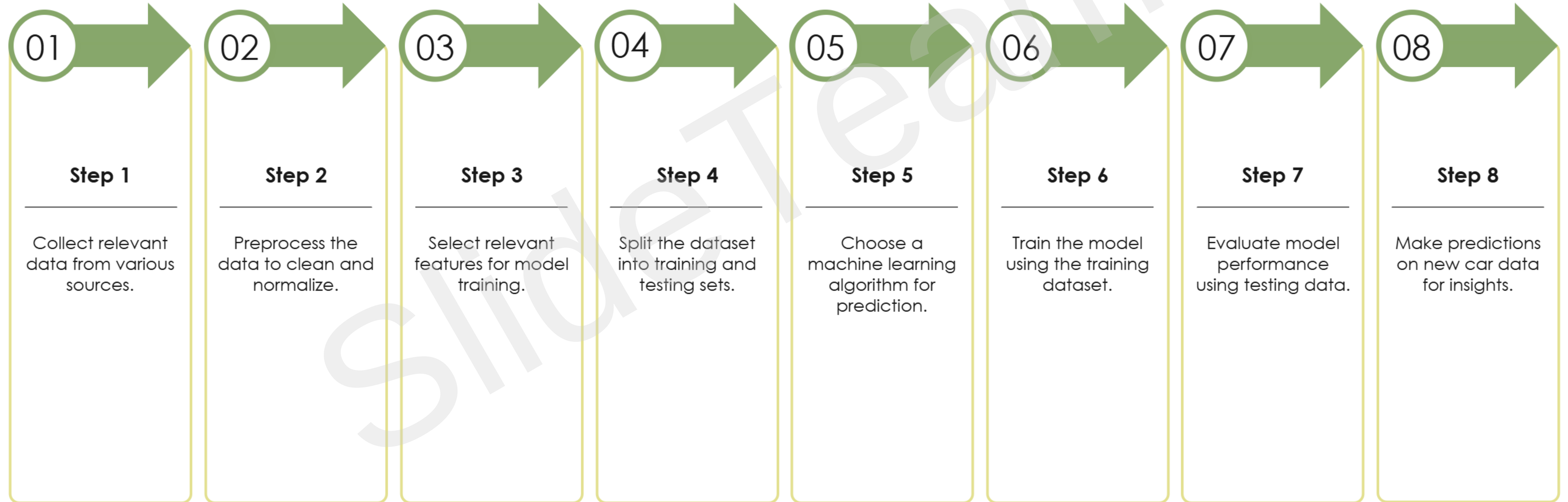


“

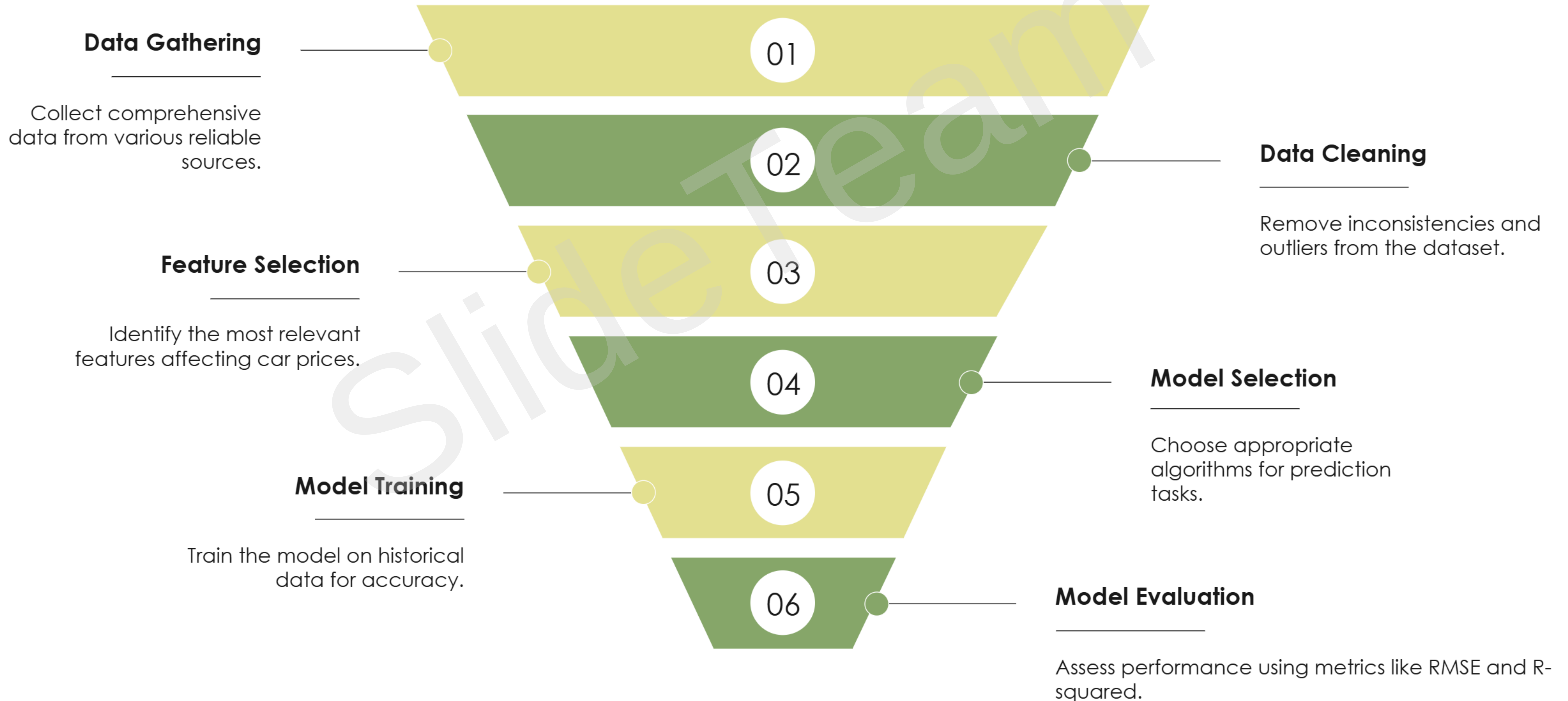
A reliable service that helped me find the best deals.

”

Hands-On Demonstration of Predictions



Strategies for Accurate Predictions



Final Thoughts and Summary

01

Importance of Data

Quality data is essential for accurate car price predictions.

03

Market Trends

Staying updated with market trends can significantly impact predictions.

02

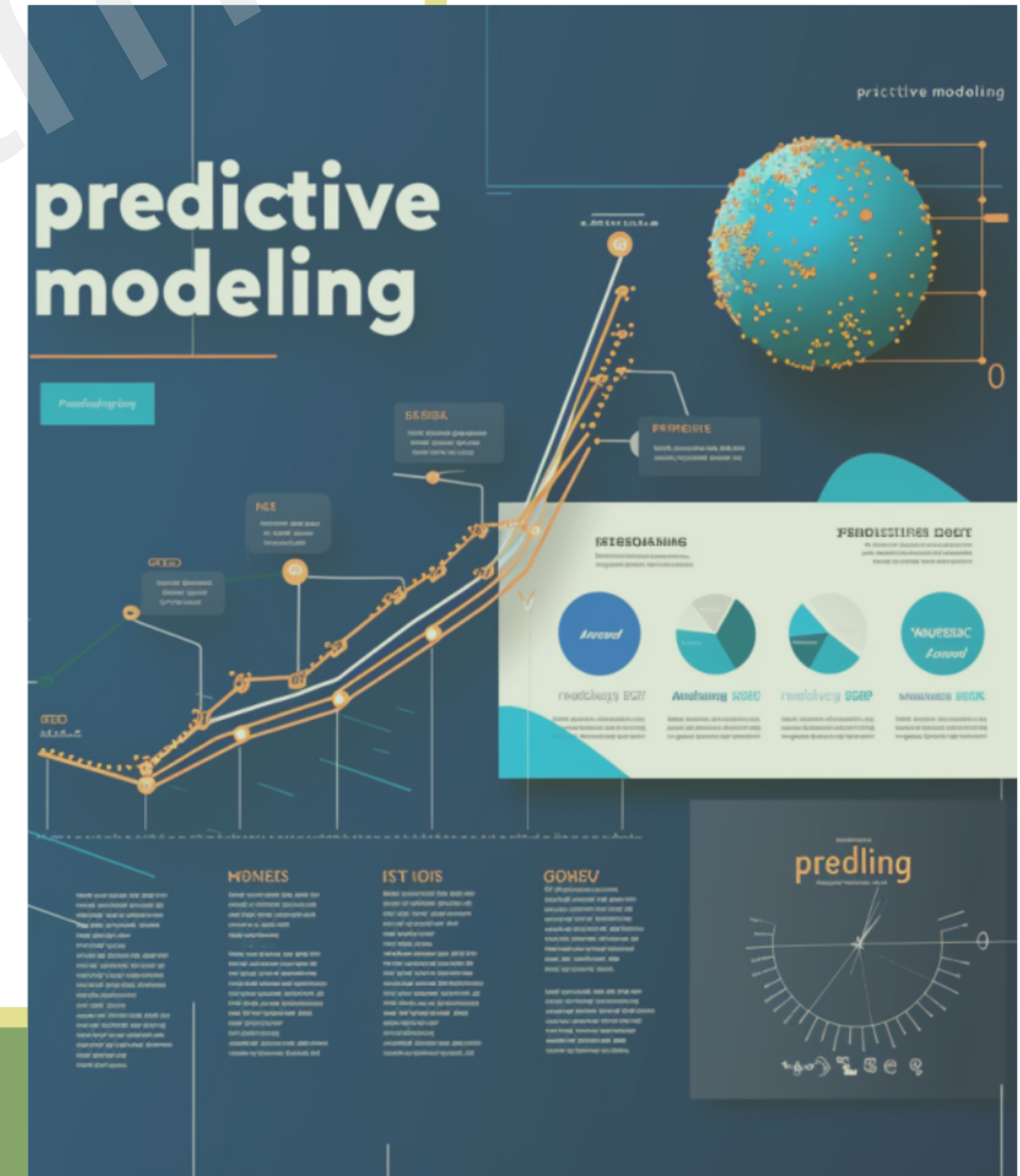
Machine Learning Models

Utilizing advanced models enhances prediction accuracy for car pricing.

04

Future Developments

Continuous research and development will lead to better predictive tools.



Thank You



Address

124/258 , Govind Nagar , kanpur



Email Address

radhikamaggo2608@gmail.com



Contact Number

(+91-8957370676

